CSP-203: Software Tools & Technologies Lab



Chapter – 3

Latex: Beamer

(Lab Session – 13)

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Contents

- Introduction
- Title page
- Table of contents
- Adding effects on presentation
- Highlighting important sentences and word
- Animation in Beamer
- Bibliography

❑ Introduction (Start of creating a presentation):

beamer.tex

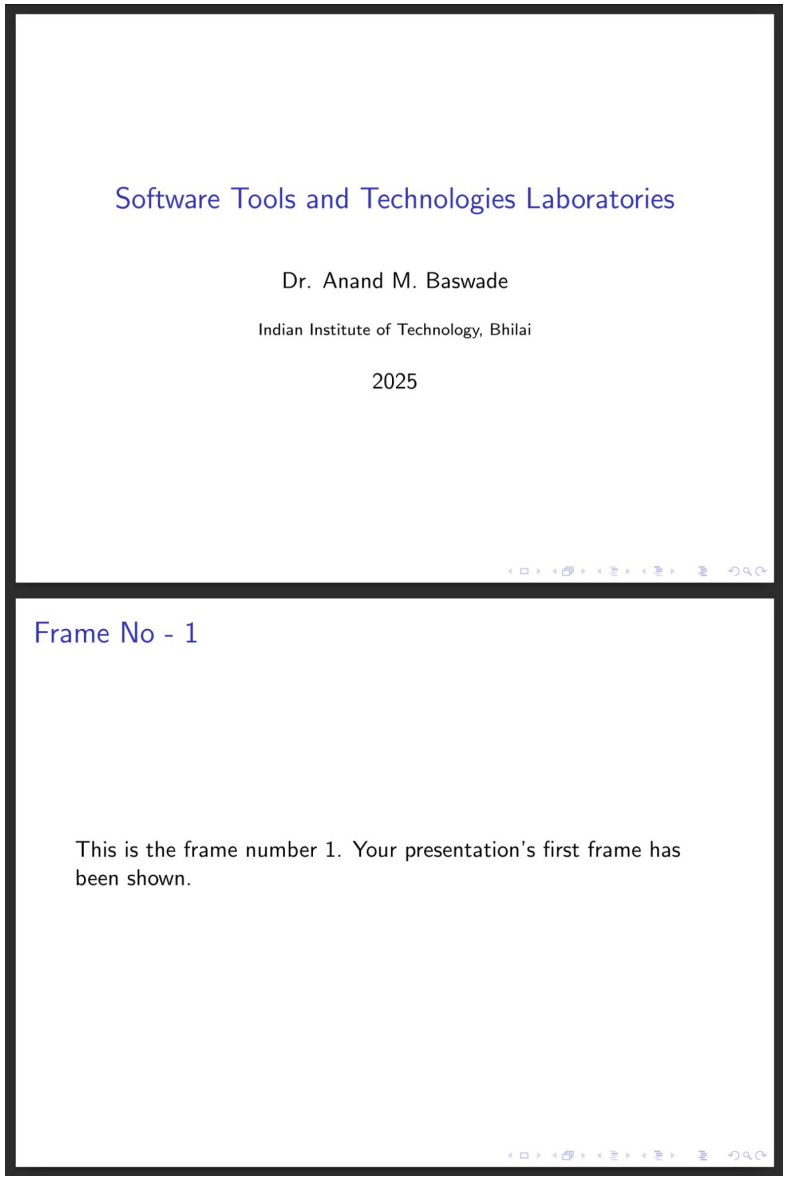
```latex
\documentclass{beamer}
\title{Software Tools and Technologies Laboratories}
\author{Dr. Anand M. Baswade}
\institute{Indian Institute of Technology, Bhilai}
\date{2025}

\begin{document}

    \frame{\titlepage}

    \begin{frame}
        \frametitle{Frame No - 1}
            This is the frame number 1.
            Your presentation's first frame has been shown.
    \end{frame}

\end{document}
```



Details into title page:

beamer.tex

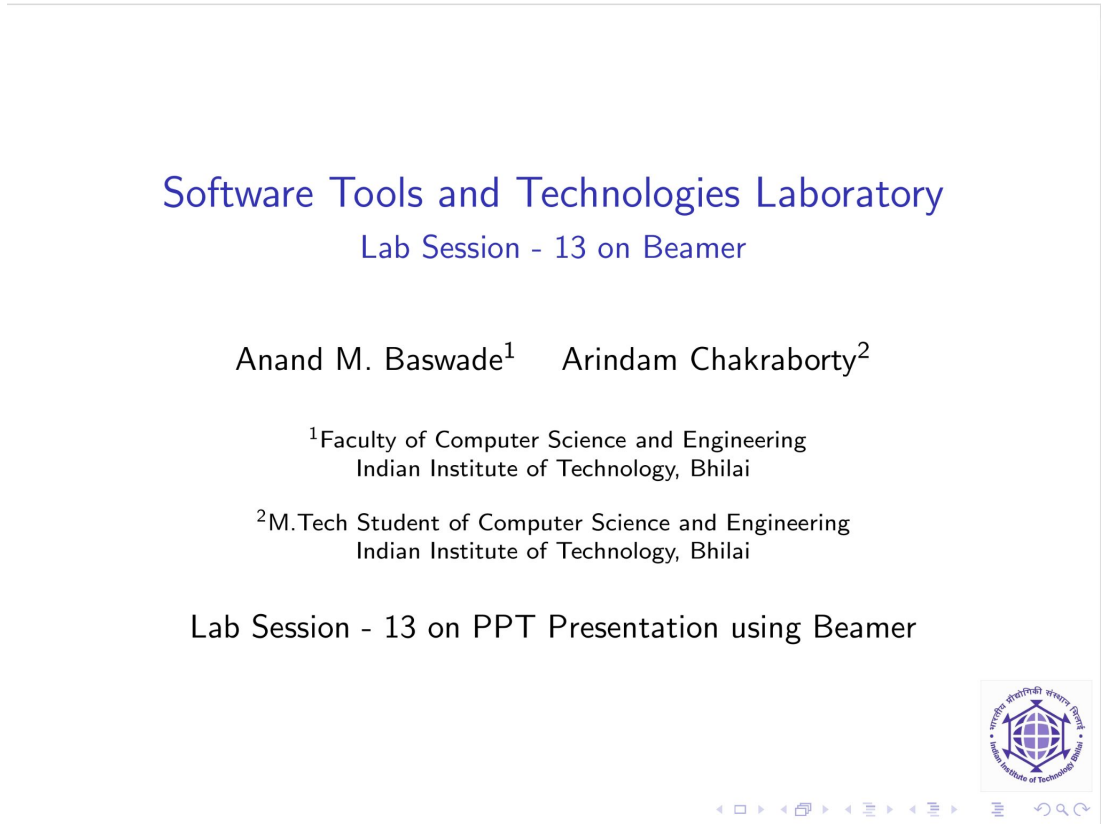
```
\documentclass{beamer}
%\usetheme{Madrid}
\title[About Beamer]{Software Tools and Technologies Laboratory}

\subtitle{Lab Session - 13 on Beamer}

\author[Anand, Arindam]{Anand M. Baswade\inst{1} \and Arindam Chakraborty\inst{2}}

\institute[IIT]
{
    \inst{1}
    Faculty of Computer Science and Engineering\\Indian Institute of Technology, Bhilai
    \and
    \inst{2}
    M.Tech Student of Computer Science and Engineering\\Indian Institute of Technology, Bhilai
}

\date[Beamer 2025]{Lab Session - 13 on PPT Presentation using Beamer}
\logo{\includegraphics[height=1.3cm]{iit-bhilai}}
```



Details into title page:

```
\documentclass{beamer}
\usetheme{Madrid}
\title[About Beamer]{Software Tools and Technologies Laboratory}

\subtitle{Lab Session - 13 on Beamer}

\author[Anand, Arindam]{Anand M. Baswade\inst{1} \and Arindam Chakraborty\inst{2}}

\institute[IIT]
{
    \inst{1}
    Faculty of Computer Science and Engineering\\Indian Institute of Technology, Bhilai
    \and
    \inst{2}
    M.Tech Student of Computer Science and Engineering\\Indian Institute of Technology, Bhilai
}

\date[Beamer 2025]{Lab Session - 13 on PPT Presentation using Beamer}
\logo{\includegraphics[height=1.3cm]{iit-bhilai}}
```

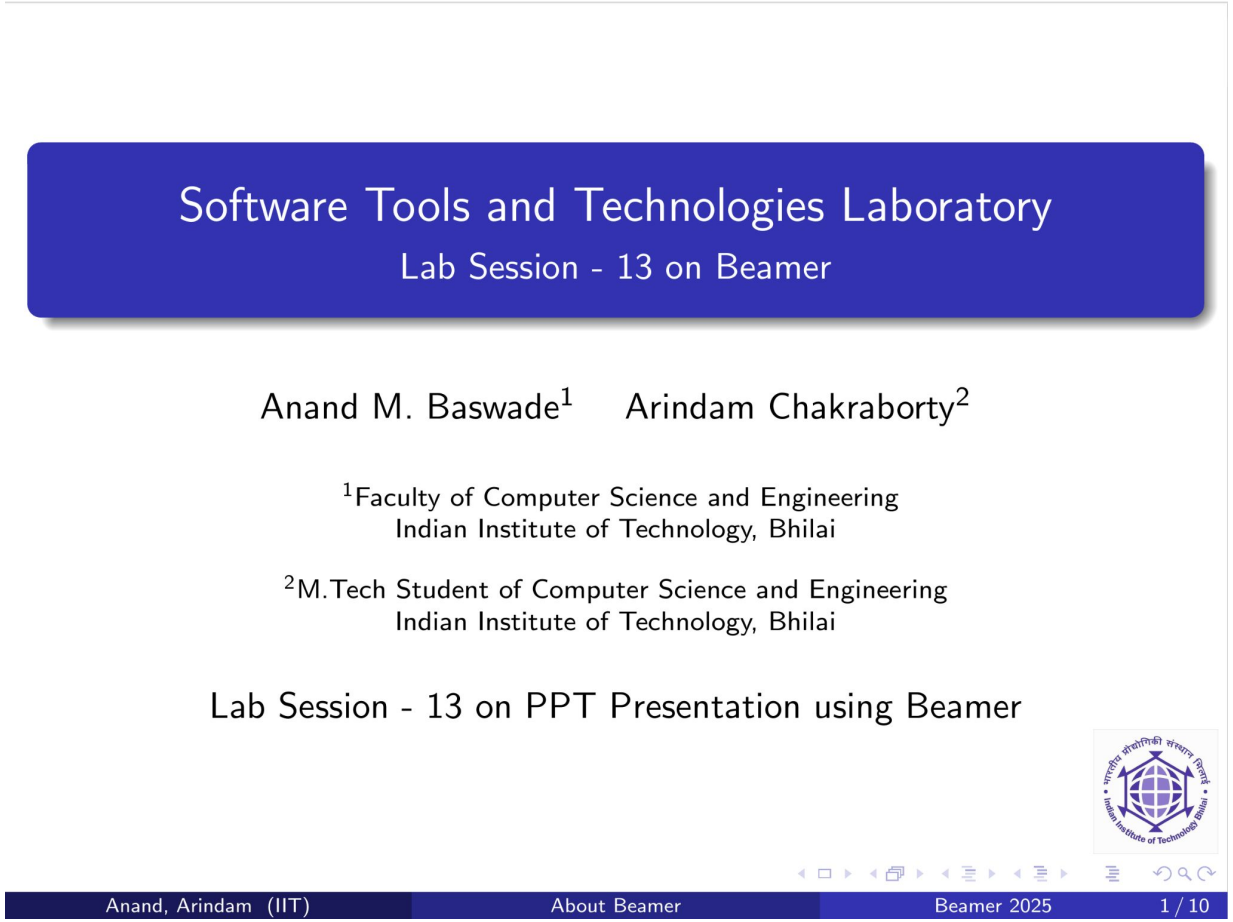


Table of contents:

```
\begin{frame}
    \frametitle{Table of Contents}
    \tableofcontents
\end{frame}
```

➔ You can add a table of contents at the beginning of the document.

```
\AtBeginSection[]
{
    \begin{frame}
        \frametitle{Table of Contents}
        \tableofcontents[currentsection]
    \end{frame}
}
```

➔ Just add the code to the preamble of your LaTeX document highlight the title of the current section.

➔ If you use \AtBeginSubsection[] instead of \AtBeginSection[], the table of contents will appear at the beginning of each subsection.

1 Introduction

2 Next Section

3 Third Section

4 Highlighting



Table of Contents

1 Introduction

2 Next Section

3 Third Section

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Hands on using beamer

This is a text in second frame. For the sake of showing an example.

- Text visible on slide 1,2,3,4

❑ Adding effects on presentation:

➔ You can make the items to appear on slides one by one.

```
\begin{itemize}
    \item<1-> Text visible on slide 1,2,3,4
    \item<2-> Text visible on slide 2,3,4
    \item<3> Text visible on slide 3 but not on slide 4
    \item<4-> Text visible on slide 4
\end{itemize}
```

```
\begin{itemize}
    \item<1-> Text visible on slide 1,2,3,4
    \item<2-> Text visible on slide 2,3,4
    \item<3-> Text visible on slide 3,4
    \item<4-> Text visible on slide 4
\end{itemize}
```

➔ You can make texts appear one after another.

```
\section{Third Section}
\begin{frame}
    \frametitle{Text Appearing One by one}
        In this slide \pause

        the text will be partially visible \pause

        And finally everything will be there
    \end{frame}
```

❑ Highlighting Important Sentences and Words:

➔ \alert{} will change the style of the word inside the braces.

➔ **\begin{block}{Remark} \end{block}**:

A block box will wrap the text in a box with the same style as the rest of the presentation. The text inside the braces after the \begin{block} code is the title of the box.

➔ **\begin{alertblock}{Important theorem} \end{alertblock}**:

The same as block but the style contrasts the one used by the presentation.

➔ **\begin{examples} \end{examples}:**

Again, is very similar to block, the box has a different style but less contrasting than alertblock.

❑ Highlighting Important Sentences and Words:

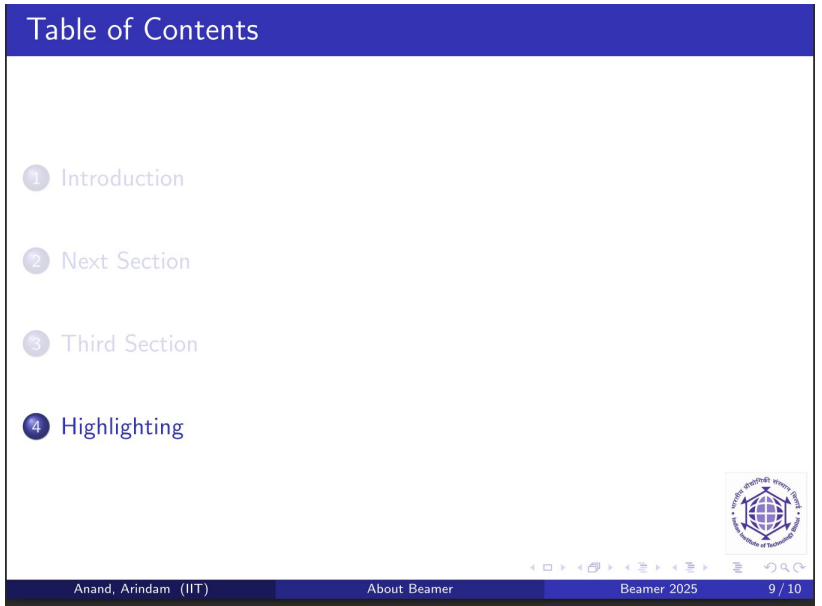
```
\section{Highlighting}
\begin{frame}
    \frametitle{Highlighting texts}

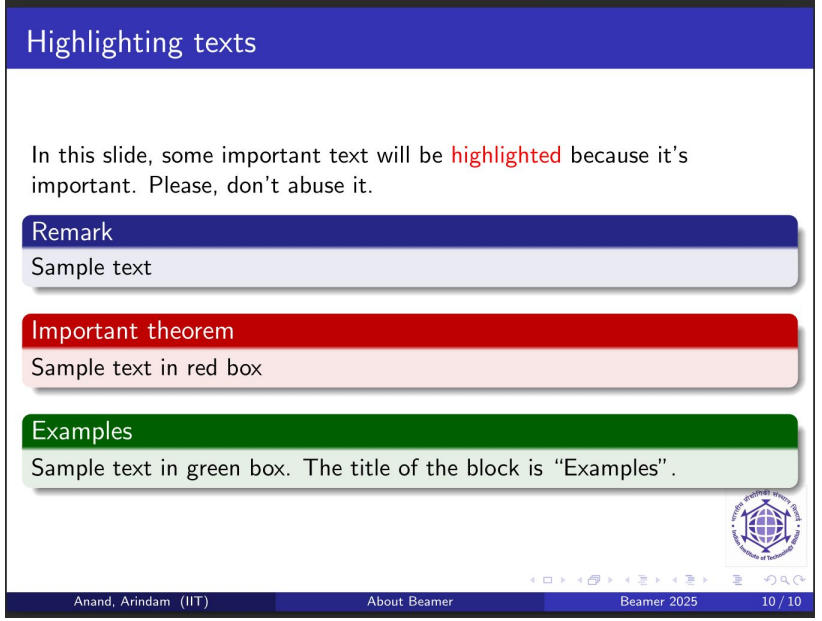
    In this slide, some important text will be
    \alert{highlighted} because it's important.
    Please, don't abuse it.

    \begin{block}{Remark}
        Sample text
    \end{block}

    \begin{alertblock}{Important theorem}
        Sample text in red box
    \end{alertblock}

    \begin{examples}
        Sample text in green box. The title of the block is ``Examples".
    \end{examples}
\end{frame}
```

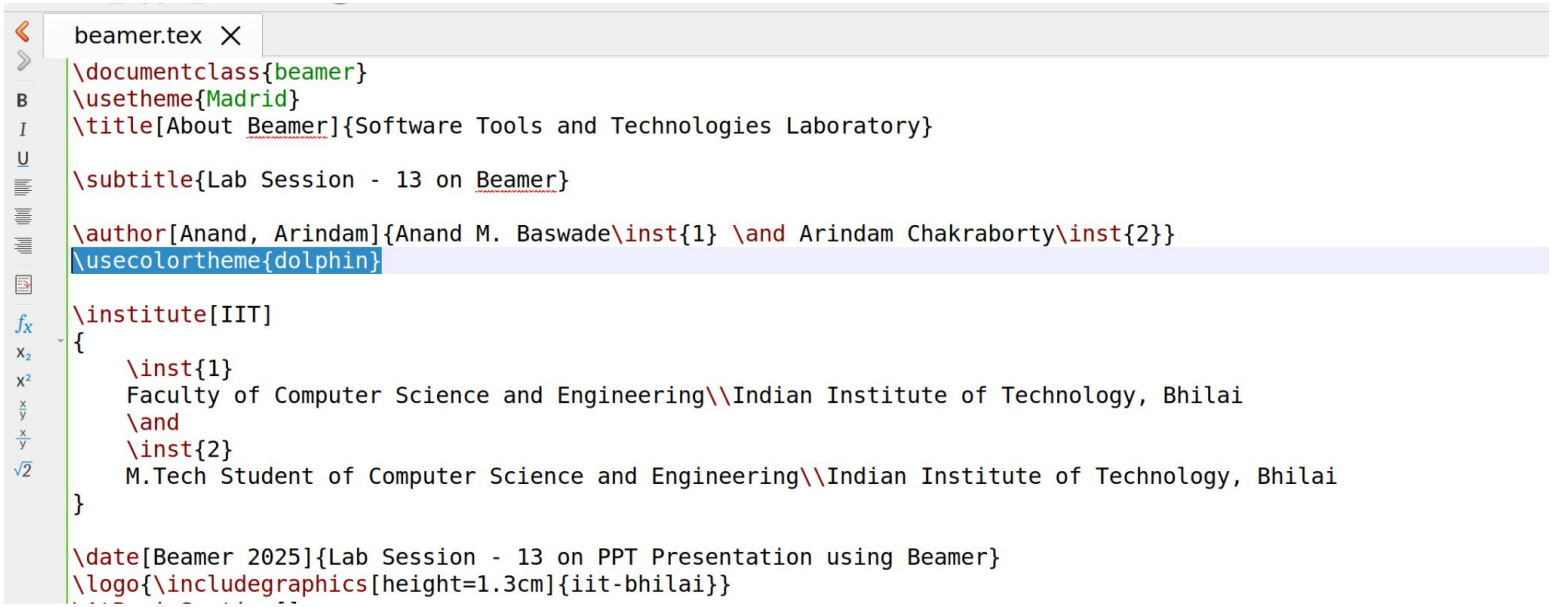




❑ Highlighting Important Sentences and Words:

➔ In LaTeX Beamer, \usecolortheme{} lets you set the color scheme. There are several built-in color themes you can use directly. Here's the list of default Beamer color themes:

- default
- albatross
- beaver
- beetle
- crane
- dolphin
- dove
- fly
- lily
- orchid
- rose
- seagull
- seahorse
- sidebartab
- structure
- whale
- wolverine



Beamer: Environment – Mathematics Blocks

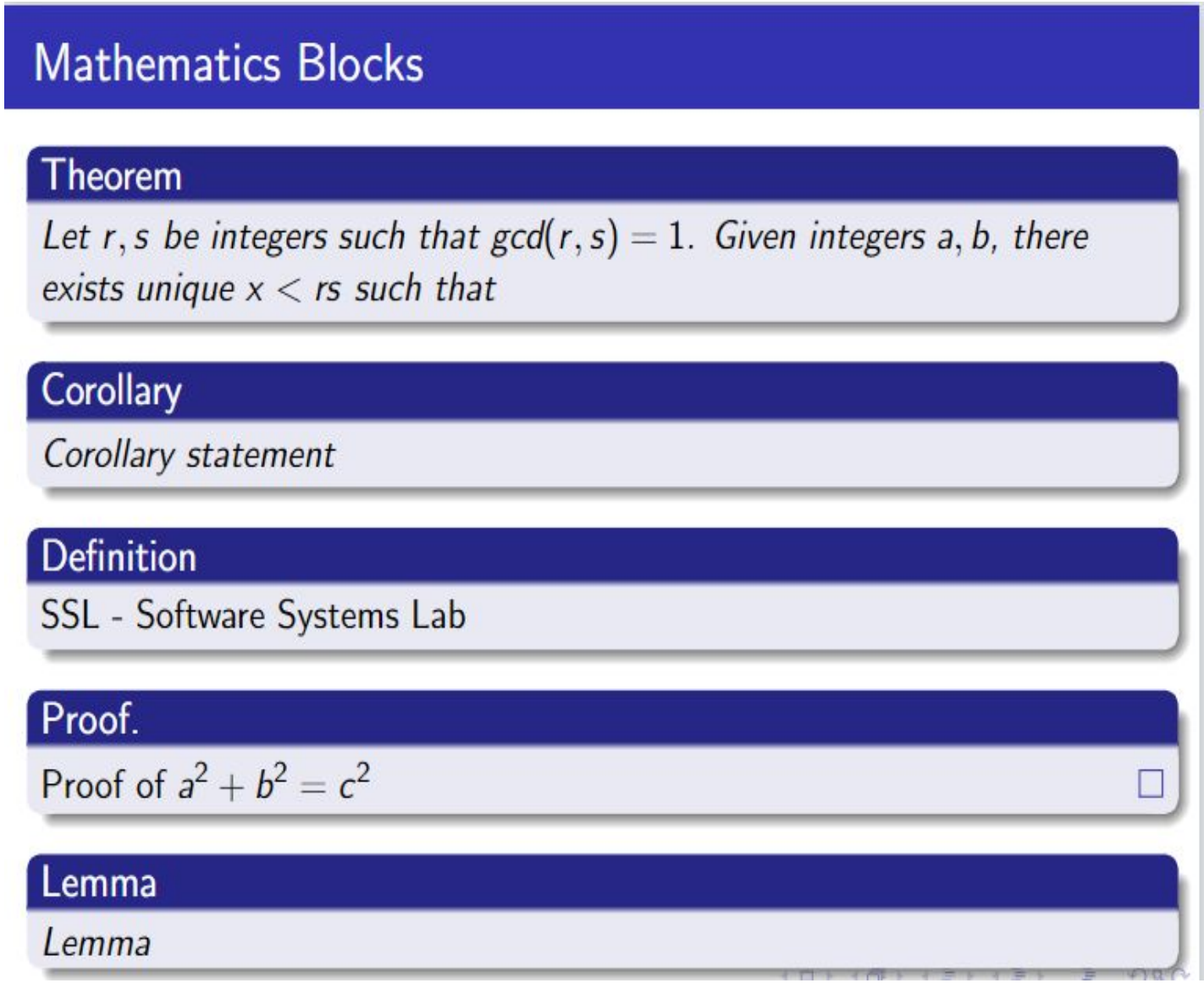
```
\begin{theorem}
Let $r, $s be integers such that gcd$(r, s)=1$.
Given integers $a,b$, there exists unique $x
<rs$ such that
\end{theorem}
```

```
\begin{corollary}
Corollary statement
\end{corollary}
```

```
\begin{definition}
SSL - Software Systems Lab
\end{definition}
```

```
\begin{proof}
Proof of $a^2 + b^2 = c^2$
\end{proof}
```

```
\begin{lemma}
Lemma
\end{lemma}
```

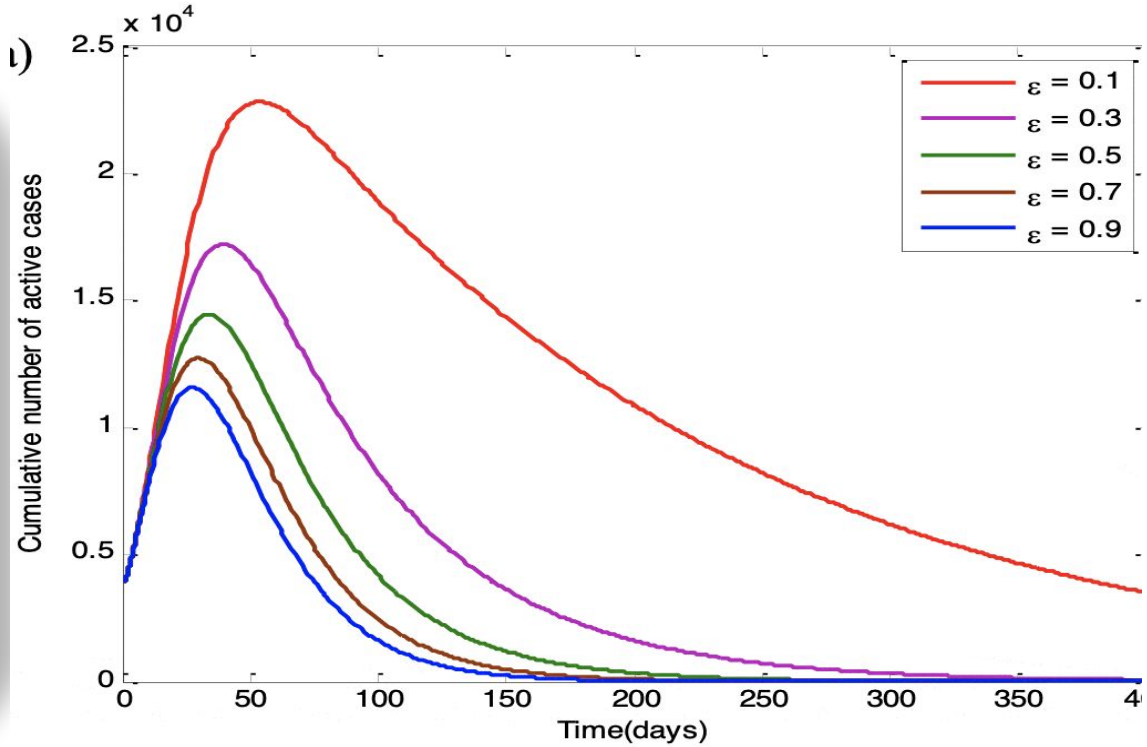


Beamer: Animations - PNG

\animategraphics[<options>]{<frames per second>}{<name without extension>}{<first frame>}{<last frame>}

```
%% Beamer-Environment
\begin{frame}{GIFs in Beamer}
\centering
\animategraphics[autoplay,loop,width=4cm]{10}{fig/effects-}
{0}{3}
\end{frame}
```

Output:



- Options: autoplay (the images play automatically), loop (once the images start moving, it continues in a loop (without stopping)
- frames per second]
- animation filename: must specify the animation filename without its extension and without the number
- first frame: the number of the first image
- last frame: number of the images

❑ Bibliography Example:

Open ref.bib ~/Desktop/TA STTL Save

```bibtex
@book{lamport1994latex,
  title        = {LaTeX: A Document Preparation System},
  author       = {Lamport, Leslie},
  year         = {1994},
  publisher    = {Addison-Wesley},
  edition      = {2}
}

@article{knuth1984tex,
  title        = {The TeXbook},
  author       = {Knuth, Donald E.},
  journal      = {Computers \& Mathematics with Applications},
  volume       = {9},
  number       = {1},
  pages        = {1--12},
  year         = {1984},
  publisher    = {Elsevier}
}

@inproceedings{goossens1994latexcompanion,
  title        = {The \LaTeX\ Companion},
  author       = {Goossens, Michel and Mittelbach, Frank and Samarin, Alexander},
  booktitle    = {Proceedings of the TeX Users Group Meeting},
  year         = {1994},
  organization = {TUG}
}
```

```latex
    \begin{frame}{Cite}
        This presentation is based on \LaTeX{} \cite{lamport1994latex},
        \TeX{} \cite{knuth1984tex}, and the \LaTeX\ Companion \cite{goossens1994latexcompanion}.
    \end{frame}
    \begin{frame}[allowframebreaks]{References}
        \bibliography{ref}
    \end{frame}
\end{document}
```

Cite

This presentation is based on LaTeX [3], TeX [2], and the LaTeX Companion [1].



References I

Michel Goossens, Frank Mittelbach, and Alexander Samarin.
The companion.
In *Proceedings of the TeX Users Group Meeting*. TUG, 1994.

Donald E. Knuth.
The texbook.
Computers & Mathematics with Applications, 9(1):1–12, 1984.

Leslie Lamport.
LaTeX: A Document Preparation System.
Addison-Wesley, 2 edition, 1994.




THANK
YOU